$$= \int_0^x \frac{x}{2}dx = \frac{x^2}{4}$$

Hence, $F(x) = \begin{cases} 0 & if\ x < 0 \\ \frac{x^2}{4} & if\ 0 < x < 2 \\ 1 & if\ x \geq 2 \end{cases}$

(ii) $P\left(\frac{1}{2} < X \leq 1\right) = F(1) - F\left(\frac{1}{2}\right)$

$$= \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$$

This probability can be computed using p.d.f

$$\int_{\frac{1}{2}}^{1} \frac{x}{2}dx = \frac{1}{2}\left(\frac{x^2}{2}\right)_{\frac{1}{2}}^{1}$$

$$= \frac{3}{16}$$

(iii) $P\ (X = 1.5) = 0$

NOTE

For a continuous random variable probability at a particular point is 0

9.5 Joint and marginal probability mass functions

In real life situations we may be observed two or more random variables on the individuals simultaneously. For instance, blood pressure and cholesterol for each individual are measured simultaneously. In such cases we require the concept of bi-variate random variable represented by (X, Y), where X and Y are univariate random variables.

9.5.1 Definition (Joint p.m.f)

Let (X, Y) be a discrete bivariate random variable. Then $p\ (x, y)$ is called the joint probability mass function of (X, Y) if the following conditions are satisfied.

$p\ (x, y) \geq 0 \quad \forall\ x, y$

$\sum_{x,y} p(x,y) = 1$

NOTE

$p\ (x, y) = p\ (X = x, Y = y)$

Definition (Marginal Probability Mass Function)

Given a joint probability mass function $p\ (x, y)$, then $p\ (x) = \sum_y p(x,y)$ is called marginal probability mass function of X. Similarly $p\ (y) = \sum_x p(x,y)$ is called the marginal probability mass function of Y.

For example,

There are 10 tickets in a bag which are numbered 1, 2, 3, ...10. Two tickets are drawn at random one after the other with replacement.